Petrogenesis and Mantle Source Characteristics of Cenozoic Basalts

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Abstract. We present high-precision major and trace element data for mantle peridotite xenoliths...

1. Introduction

The subcontinental lithospheric mantle beneath eastern China has experienced profound reactivation.

Analytical techniques including electron microprobe and MC-ICP-MS were employed at standard conditions.

Table 1. Representative major oxide (wt%) and trace element compositions of mantle olivine and clinopyroxene.

Sample	Type	SiO2	TiO2	Al2O3	FeO	MnO	MgO	CaO	Na2O	Mg#	87Sr/86Sr	143Nd/144Nd
SMP-01	Ol	40.95 ± 0.12	0.20	3.75 ± 0.08	8.32	0.16	37.90 ± 0.15	1.50	0.36	89.0	0.70358 ± 0.00004	0.512765 ± 0.000012
SMP-02	Cpx	41.40 ± 0.12	0.28	4.10 ± 0.08	8.54	0.18	37.60 ± 0.15	2.15	0.47	88.9	0.70366 ± 0.00004	0.512780 ± 0.000012
SMP-03	Ol	41.85 ± 0.12	0.36	4.45 ± 0.08	8.76	0.20	37.30 ± 0.15	2.80	0.58	88.8	0.70374 ± 0.00004	0.512795 ± 0.000012
SMP-04	Cpx	42.30 ± 0.12	0.44	4.80 ± 0.08	8.98	0.22	37.00 ± 0.15	3.45	0.69	88.6	0.70382 ± 0.00004	0.512810 ± 0.000012
SMP-05	Ol	42.75 ± 0.12	0.52	5.15 ± 0.08	9.20	0.24	36.70 ± 0.15	4.10	0.80	88.5	0.70390 ± 0.00004	0.512825 ± 0.000012
SMP-06	Cpx	43.20 ± 0.12	0.60	5.50 ± 0.08	9.42	0.26	36.40 ± 0.15	4.75	0.91	88.3	0.70398 ± 0.00004	0.512840 ± 0.000012
SMP-07	Ol	43.65 ± 0.12	0.68	5.85 ± 0.08	9.64	0.28	36.10 ± 0.15	5.40	1.02	88.2	0.70406 ± 0.00004	0.512855 ± 0.000012
SMP-08	Cpx	44.10 ± 0.12	0.76	6.20 ± 0.08	9.86	0.30	35.80 ± 0.15	6.05	1.13	88.0	0.70414 ± 0.00004	0.512870 ± 0.000012
SMP-09	Ol	44.55 ± 0.12	0.84	6.55 ± 0.08	10.08	0.32	35.50 ± 0.15	6.70	1.24	87.9	0.70422 ± 0.00004	0.512885 ± 0.000012
SMP-10	Cpx	45.00 ± 0.12	0.92	6.90 ± 0.08	10.30	0.34	35.20 ± 0.15	7.35	1.35	87.7	0.70430 ± 0.00004	0.512900 ± 0.000012
SMP-11	Ol	45.45 ± 0.12	1.00	7.25 ± 0.08	10.52	0.36	34.90 ± 0.15	8.00	1.46	87.5	0.70438 ± 0.00004	0.512915 ± 0.000012
SMP-12	Cpx	45.90 ± 0.12	1.08	7.60 ± 0.08	10.74	0.38	34.60 ± 0.15	8.65	1.57	87.4	0.70446 ± 0.00004	0.512930 ± 0.000012
SMP-13	Ol	46.35 ± 0.12	1.16	7.95 ± 0.08	10.96	0.40	34.30 ± 0.15	9.30	1.68	87.2	0.70454 ± 0.00004	0.512945 ± 0.000012
SMP-14	Cpx	46.80 ± 0.12	1.24	8.30 ± 0.08	11.18	0.42	34.00 ± 0.15	9.95	1.79	87.1	0.70462 ± 0.00004	0.512960 ± 0.000012
SMP-15	Ol	47.25 ± 0.12	1.32	8.65 ± 0.08	11.40	0.14	33.70 ± 0.15	10.60	1.90	87.0	0.70470 ± 0.00004	0.512975 ± 0.000012
SMP-16	Cpx	47.70 ± 0.12	1.40	9.00 ± 0.08	11.62	0.16	33.40 ± 0.15	11.25	2.01	86.8	0.70478 ± 0.00004	0.512990 ± 0.000012
SMP-17	Ol	48.15 ± 0.12	1.48	9.35 ± 0.08	11.84	0.18	33.10 ± 0.15	11.90	2.12	86.7	0.70486 ± 0.00004	0.513005 ± 0.000012
SMP-18	Cpx	48.60 ± 0.12	1.56	9.70 ± 0.08	12.06	0.20	32.80 ± 0.15	12.55	2.23	86.5	0.70494 ± 0.00004	0.513020 ± 0.000012
SMP-19	Ol	49.05 ± 0.12	1.64	10.05 ± 0.08	12.28	0.22	32.50 ± 0.15	13.20	2.34	86.4	0.70502 ± 0.00004	0.513035 ± 0.000012
SMP-20	Cpx	49.50 ± 0.12	1.72	10.40 ± 0.08	12.50	0.24	32.20 ± 0.15	13.85	0.35	86.2	0.70510 ± 0.00004	0.513050 ± 0.000012
SMP-21	Ol	49.95 ± 0.12	1.80	10.75 ± 0.08	8.22	0.26	31.90 ± 0.15	14.50	0.46	86.0	0.70518 ± 0.00004	0.513065 ± 0.000012
SMP-22	Cpx	50.40 ± 0.12	1.88	11.10 ± 0.08	8.44	0.28	31.60 ± 0.15	15.15	0.57	85.9	0.70526 ± 0.00004	0.513080 ± 0.000012
SMP-23	Ol	50.85 ± 0.12	0.16	11.45 ± 0.08	8.66	0.30	31.30 ± 0.15	15.80	0.68	85.8	0.70534 ± 0.00004	0.513095 ± 0.000012
SMP-24	Cpx	51.30 ± 0.12	0.24	11.80 ± 0.08	8.88	0.32	31.00 ± 0.15	16.45	0.79	85.6	0.70542 ± 0.00004	0.513110 ± 0.000012

Notes: FeO is total iron as FeO. Mg# = 100 × Mg / (Mg + Fe) atomic. Analytical uncertainties (± 1·) reported for isotopic ratios.

2. Discussion and Geochemical Implications

The Mg# values correlate strongly with compatible trace element abundances, indicating melt extraction.

References

1. McDonough, W.F. & Sun, S.-s. (1995) Chem. Geol. 120, 223·253.